

Notes (Week 1 Monday)

These are the notes I wrote during the lecture.

Q: why not RFC:s?

A: it may be ambiguous
it's written in text, you could
misinterpret it

Informally,
a set is a collection of elements

$x \in S$	"x in S"	x is a member (element) of the set S
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$x \notin S$ "x not in S" abbreviates $\neg(x \in S)$

- ¬ negation

$x \in A \cap B$ "x in (A and B)" intersection

If A and B are sets,

$A \cap B$ is a set that contains all elements common to A and B .

`{1,2,3}` explicitly enumerated elements of a set

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"the set containing only the elements 1, 2 and 3"
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$$\{1, 2, 3\} \cap \{2, 3, 4\} = \{2, 3\}$$

Q: Subset is not like the others

A: It's asymmetrical

$$A \cup B = B \cup A \quad (\text{commutativity})$$

$$A \cap B = B \cap A \quad (\text{commutativity})$$

$$A \subseteq B \quad \neq \quad B \subseteq A \quad (\text{not symmetry})$$

(only holds if A, B are equal)

A: The operator doesn't return a set

The "types" are different:

$$U : \text{set} \times \text{set} \rightarrow \text{set}$$

$$n : \text{set} \times \text{set} \rightarrow \text{set}$$

$\subseteq : \text{set} \times \text{set} \rightarrow \text{bool}$

Union and intersection are *combinators* of sets:

if I have two sets, I can make a new set

Subset inclusion is a *predicate* about sets

makes a truth claim about two sets, but
isn't itself a set.

Q: Brackets? Order of operations:

wrt $\subseteq$ and $\cup, \cap$

$A \subseteq A \cup B$ parses as $A \subseteq (A \cup B)$

the "types" don't make sense otherwise

$(A \subseteq A) \cup B$ "doesn't typecheck"
^^^^ ^
bool set

Theorem union_subset:

$A \subseteq A \cup B$

Proof

By definition of $\subseteq$, it suffices to prove

$x \in A$ implies $x \in A \cup B$

By definition of union, it suffices to prove

$x \in A$ implies $(x \in A \text{ or } x \in B)$

This holds by disjunction introduction.

QED

Disjunction introduction?

A proof rule:

If A holds then $A \vee B$ holds

Set enumeration:

$\{1, 2, 3\}$

Set comprehension:

$\{x \mid P\}$ or $\{x : P\}$ or $\{x . P\}$

Here, P is a predicate (truth claim about x)

$\{x \mid P\}$ "the set of all x such that P holds"

Q: is zero a natural number?

A: yes

$\{x \mid x \in \mathbb{N} \text{ and } x < 5\} = \{0,1,2,3,4\}$

$x = \{1,2\}$

$\{y \mid y \subseteq \{1,2\}\} = \{\{\},$
 $\{1\},$
 $\{2\},$
 $\{1,2\}\}$

powerset

$\{y \mid y \in x\}$ "the set of all elements of x "

We're taking the view that if two sets have the same elements, they're equal. (principle of extensionality)

$x \in x$ "x contains itself as an element"

only possible if x has "infinite depth"

Most mathematicians start courses by saying that the foundations we use are

ZFC (Zermelo–Fraenkel set theory with the axiom of Choice)

ZFC has an axiom called the axiom of foundation, which says that infinite depth sets can't exist

Suppose $y \in y$

By definition, $y \in \{x \mid x \notin x\}$

Which implies that the predicate in the comprehension holds about y itself, i.e.

$y \notin y$

Q:Why is this not just a minor flaw?

A:If there's a contradiction somewhere, it's everywhere

Most logics allow *proof by contradiction*

In order to prove P , it suffices that from $\neg P$, False follows.

If our system allows Russell's paradox, we can prove False from nothing.

Therefore, we can prove everything.

We want to build a logical system that is *consistent*

Consistent: some things are unprovable

Here, we distinguished between constant symbols
and function symbols.

We don't have to:

constant symbols are essentially function symbols
with arity 0

We haven't specified precisely what functions
and constant symbols exist, nor will we.

These are not part of the logic as such,
they are the things the logic is talking about.
Instead, the logic is parameterised on these.

$\forall n. n! = n \leftrightarrow (n=1 \vee n=3)$ "for all n, ..."

If I ask whether this is true

$n! = n \leftrightarrow (n=1 \vee n=3)$

The answer is: it depends on the value of n.

Formulas on their are neither true nor false (in general)

Instead, their truth value depends on a **valuation**
it depends on what values we want to give to the variables

$\{x \mid x \notin x\}$ a set comprehension
is a set defined in terms of a predicate
a term defined in terms of a predicate

$l \leq m < h$

At the else statement, if we reach it,
therefore the following must hold:

$\neg(l \leq m < h)$
 $\equiv$ (by arithmetic)
 $\neg(l \leq m \wedge m < h)$
 $\equiv$ (by de Morgan's laws $\neg(A \wedge B) \equiv \neg A \vee \neg B$)
 $\neg(l \leq m) \vee \neg(m < h)$
 $\equiv$ (by arithmetic)
 $l > m \vee m \geq h$

Does this equivalence hold?

$\neg(l \leq m < h) \wedge m < l$
 $\equiv$
 $\neg(l \leq m < h) \wedge \neg(m \geq h)$